

TULLIBY LAKE AGCM, AB, Canada

WMO: 718570

Lat: 53.6644N

Lon: 110.0810W

Elev: 596

StdP: 94.37

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-33.4	-30.5	-37.2	0.1	-33.3	-34.1	0.2	-30.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	27.9	17.2	26.1	16.8	24.6	16.1	19.4	25.3	18.2	23.9	17.2	22.6	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	13.2	22.4	16.0	12.2	21.0	14.9	11.4	19.9	57.9	25.5	54.0	24.1	50.6	22.7	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-38.3	31.0	2.1	1.6	-39.8	32.2	-41.0	33.2	-42.2	34.1	-43.7	35.3		
(5)			-38.4	21.9	2.1	1.5	-39.9	23.0	-41.1	23.9	-42.3	24.8	-43.8	25.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.9	-13.5	-14.2	-6.4	2.5	10.8	15.0	17.1	15.8	11.0	3.6	-6.1	-14.0
	DBStd	13.11	8.62	7.89	7.67	5.35	4.48	3.11	2.59	3.04	4.46	4.78	6.87	8.21
	HDD10.0	3661	728	676	508	230	46	4	0	1	39	204	482	743
	HDD18.3	6031	986	909	766	476	236	105	54	87	222	456	732	1002
	CDD10.0	709	0	0	0	4	70	155	221	182	69	7	0	0
	CDD18.3	36	0	0	0	0	2	6	17	10	2	0	0	0
	CDH23.3	624	0	0	0	2	67	104	205	167	76	2	0	0
	CDH26.7	107	0	0	0	0	9	17	34	28	18	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	446	18	12	20	22	45	81	85	64	42	15	18	22
	PrecMax	577	38	28	42	58	142	144	203	155	82	40	43	51
	PrecMin	288	7	1	1	1	8	26	11	7	6	3	2	5
	PrecStd	85	8	7	11	16	33	34	40	37	22	10	12	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	4.3	11.5	22.3	27.9	28.9	29.8	29.6	29.3	21.7	11.0	4.7
		MCWB	2.0	1.7	5.9	10.0	13.9	16.2	19.7	18.8	16.9	12.5	5.8	1.6
	2%	DB	3.2	1.8	7.9	17.7	25.4	26.2	27.3	26.9	25.4	16.8	7.0	1.9
		MCWB	1.0	0.0	3.5	8.2	12.8	15.8	18.8	18.1	15.0	9.9	3.6	-0.2
	5%	DB	1.4	-0.1	4.9	14.5	22.4	24.1	25.5	25.1	22.6	14.0	4.1	-0.5
		MCWB	-0.2	-1.3	2.0	6.6	11.9	15.0	17.9	17.2	14.2	8.4	1.9	-2.1
	10%	DB	-1.0	-2.1	3.0	11.4	20.2	22.2	24.0	23.4	19.4	11.2	2.2	-2.8
		MCWB	-2.4	-3.3	0.7	5.0	10.7	13.9	16.9	16.5	12.6	6.9	0.7	-3.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	1.9	6.4	10.2	14.9	18.4	21.7	20.9	17.8	12.9	6.5	1.9
		MCDB	5.1	4.1	11.1	20.7	23.5	25.1	27.2	26.1	27.1	20.5	9.8	4.4
	2%	WB	1.2	0.1	3.7	8.5	13.8	17.2	19.8	19.3	16.0	10.7	4.1	-0.2
		MCDB	2.9	1.3	7.1	16.9	22.2	23.7	25.8	25.1	24.0	15.9	6.9	1.8
	5%	WB	-0.2	-1.4	2.1	6.9	12.7	16.2	18.7	18.0	14.6	8.8	2.1	-2.0
		MCDB	1.5	0.0	4.5	14.0	20.8	21.8	24.2	23.4	21.3	13.2	3.6	-0.6
	10%	WB	-2.3	-3.4	0.8	5.5	11.5	15.2	17.6	16.9	13.1	7.1	0.6	-3.9
		MCDB	-0.9	-2.0	2.9	10.7	19.0	20.4	22.7	22.2	18.8	11.1	2.1	-2.8
Mean Daily Temperature Range	5% DB	MDBR	9.6	11.0	10.3	11.1	13.4	11.7	11.7	12.5	13.0	10.5	8.4	8.7
		MCDBR	10.8	12.1	11.8	16.8	17.0	15.1	14.5	15.7	18.7	15.5	10.7	11.1
		MCWBR	9.1	10.6	8.3	8.9	7.2	6.5	7.3	7.9	9.2	8.9	7.6	9.3
	5% WB	MCDBR	10.6	11.7	11.3	16.2	15.2	12.7	13.4	14.0	17.3	14.4	9.9	10.8
		MCWBR	9.1	10.3	8.0	8.9	6.9	6.4	7.1	7.6	8.9	8.7	7.3	9.2
Clear-Sky Solar Irradiance	taub	0.239	0.288	0.284	0.326	0.348	0.366	0.364	0.371	0.310	0.275	0.250	0.214	
	taud	2.329	2.193	2.372	2.390	2.373	2.343	2.377	2.334	2.498	2.566	2.450	2.358	
	Ebn at Noon	763	804	909	907	899	882	877	847	872	843	752	740	
	Edh at Noon	55	88	92	104	112	117	111	110	82	61	49	43	
All-Sky Solar Radiation	RadAvg	0.95	2.01	3.52	4.69	5.79	5.88	6.06	4.94	3.47	2.00	0.94	0.69	
	RadStd	0.10	0.14	0.27	0.41	0.45	0.42	0.36	0.38	0.38	0.31	0.11	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page